

Optical Networking homework #4

1a Benefit & cost of using a modulation scheme that encodes multiple bits symbol (eg., 4-ASK, 16-QAM)

Advantage/ Benefits:

Multi-level modulation encodes more bits per symbol, which increases the data rate for a fixed bandwidth.

Cost:

Reduce noise tolerance $\rightarrow$ higher SNR

Since the amplitude signal levels are closer, it becomes more susceptible to noise and distortion.

1b) Why direct modulation of an optical source more distortion problems than external modulation.

Direct modulation means $\rightarrow$ it varies the driving current of the laser / LED to encode the signal intensity.

By changing the current, changes the carrier density that alters refractive index of the medium which shifts the optical frequency (Chirp).

- The frequency chirp broadens spectrum of each pulse.

- In optical fiber, chromatic dispersion causes different frequencies to travel at slightly different speeds.

- Therefore, a chirped signal suffers more pulse spreading & inter-symbol interference (ISI) over distance.

But, external modulation keeps the laser output frequency constant, so it produces less chirp and is therefore less affected by dispersion.

- 1c) Why we prefer to modulate the current instead of the voltage when directly modulating LED's or laser diodes.

The optical o/p power of an LED/laser diode is directly proportional to the injection current above threshold not the applied voltage.

- The I-V curve is highly non-linear.
- Small changes in voltage can cause large, uncontrolled changes in current, thus making voltage modulation unstable and difficult to control.
- Current modulation gives linear, predictable control over optical o/p power & avoids overdriving or damaging the device.

- 2.) Why a higher SNR results in a lower BER.

SNR (Signal - to - noise ratio) is the signal power to the noise power ratio at the receiver.

A high SNR denotes the signals levels are much larger and more distinct compared to the random noise fluctuations added during transmission or detection process.

- a) Signal and noise power perspective:

- The receiver must decide whether each bit represents a logical 0 or 1.

- Noise adds random variations (amplitude, phase or intensity) to the received signal.

- When the signal power is much higher than the noise power (high SNR), these fluctuations don't push the signal past the decision threshold, so bits are detected correctly most of the time.

- When signal power is low to the noise power, noise can distort or cross the threshold, causing bit errors.

b) Receiver function / detection process

- Optical receivers (like photodiodes) convert optical power to electrical current.

- If the received '1' & '0' current levels overlap because of noise (shot noise, thermal noise etc), the receiver may misinterpret them.

- Increasing SNR widens the gap between these two current levels, reducing overlap & thus lowering the probability of error.

c) Eye diagram perspective

- In digital communications, an eye diagram visually shows signal quality.

- At high SNR, the eye is wide open; the top & bottom of the eye pattern are clearly separated and the crossing points are sharp and stable.

- At low SNR, noise causes amplitude variations, closing eye, the receiver has a smaller safe "decision window" increasing the likelihood of sampling errors.

3) Given parameters.

- Photodetector responsivity, $R = 5 \text{ A/W}$
- Temperature $T = 300 \text{ K}$
- Load Resistor $R_L = 10 \text{ k}\Omega$
- Noise figure $F_n = 3$ (linear)
- preamp $P_o = 0$
- Elementary Charge $q = 1.602 \times 10^{-19} \text{ C}$
- Boltmann Constant $k = 1.381 \times 10^{-23} \text{ J/K}$

$$\text{SNR} = \frac{P_s}{P_N} = \frac{(R P_{\text{opt}} G)^2}{\sigma_n^2}$$

Bandwidth Δf :
$$\Delta f = \frac{(R P_i)^2 R_L}{F_n 4 k T \text{ SNR}}$$

$$R P_i = 5 \times 10^{-7} = 5 \times 10^{-7} \text{ A}$$

$$(R P_i)^2 R_L = (5 \times 10^{-7})^2 \cdot 1 \times 10^4 = 2.5 \times 10^{-9}$$

$$4 k T F_n \cdot \text{SNR} = 4 \cdot 1.381 \times 10^{-23} \cdot 300 \cdot 3 \cdot 10 \approx 4.97 \times 10^{-19}$$

$$\Delta f = \frac{2.5 \times 10^{-9}}{4.97 \times 10^{-19}} \approx 5.03 \times 10^9 \text{ Hz}$$

$$\approx 5 \text{ GHz}$$

$$\Delta f \approx 5 \text{ GHz}$$

$$b) \sigma_I^2 = \frac{4k_B T (\Delta f)}{R_L} F_n$$

$$R = 5 \text{ A/W}$$

$$\Delta f = 1 \text{ GHz}$$

$$T = 300 \text{ K}$$

$$R_L = 10 \text{ k}\Omega$$

$$F_n = 3$$

$$P_o = 0$$

$$\text{BER} = 10^{-9} \rightarrow Q = 6$$

$$\cdot 4kT = 4 \times 1.381 \times 10^{-23} \times 300 \approx 1.6572 \times 10^{-20}$$

$$\cdot \Delta f \cdot F_n = 1.6572 \times 10^{-20} \times 1 \times 10^9 \times 3 = 4.9716 \times 10^{-11}$$

$$\cdot \text{Divide by } R_L = \frac{4.9716 \times 10^{-11}}{10^4} = 4.9716 \times 10^{-15}$$

$$\cdot \sigma_I = \sqrt{(4.9716 \times 10^{-15})} \approx 7.05 \times 10^{-8} \text{ A}$$

$$Q = \frac{I_1 - I_0}{\sigma_I - \sigma_0} \quad \& \text{ assuming } \sigma_I = \sigma_0 = \sigma_I$$

$$Q = \frac{I_1 - I_0}{2\sigma_I} = \frac{2RP_{\text{opt}}}{2\sigma_I} = \frac{RP_{\text{opt}}}{2\sigma_I}$$

$$P_{\text{opt}} = \frac{Q\sigma_I}{R} = \frac{6 \cdot 7.05 \times 10^{-8}}{5} = 8.46 \times 10^{-8} \text{ W}$$

3. Compute P_i

$$P_i = 2 \times P_{opt}$$

$$P_i = 2 \cdot 8.46 \times 10^{-8} \approx 1.69 \times 10^{-7} \text{ W} = 169 \text{ nW.}$$

$$P_i = 169 \text{ nW.}$$

c)
$$I'_s = R \cdot G \cdot P_i = 5 \cdot 10 \cdot 1.69 \times 10^{-7}$$

$$= 84.5 \times 10^{-7} \text{ A.}$$

$$= 8.45 \times 10^{-6} \text{ A}$$

~~ASE Noise Variance.~~

SNR with only thermal noise.

$$SNR_{\text{new}} =$$

Optical pre-amp noise

$$\sigma_A^2 = 4R_P^2 G(G-1) \bar{P}_{opt} (\Delta F) (n_{sp} h c / \lambda)$$

Assume $P_D = P_{opt}$

(i) $G(G-1) \approx 10 \cdot 9 = 90.$

(ii) $4R^2 P_D^2 = 4 \cdot 5^2 \cdot (1.69 \times 10^{-7})^2 = 100 \cdot (2.8561 \times 10^{-14})$

$$= 2.8561 \times 10^{-12}$$

(iii) $G(G-1) \cdot 4R^2 P_D^2 = 2.8561 \times 10^{-12} \cdot 90$

$$= 2.570 \times 10^{-10}$$

$$(iv) \quad 4R^2 P_o^2 G(G-1) \cdot P_{opt} \Delta F = 1.69 \times 10^{-7} \times 10^9 \\ = 0.169.$$

$$(v) \quad \Rightarrow 2.5705 \times 10^{-10} \cdot 0.169 \approx 4.34 \times 10^{-11} \\ \times \frac{n_{sp} h c}{\lambda} = 2.626 \times 10^{-34} \cdot 3 \times 10^8 / 1.55 \times 10^{-6} \\ = \frac{3.9756 \times 10^{-25}}{1.55 \times 10^{-6}} \approx 2.565 \times 10^{-19}.$$

$$= 4.34 \times 10^{-11} \cdot 2.565 \times 10^{-19} \approx 1.11 \times 10^{-29} A^2.$$

$\sigma_A^2 \approx 1.11 \times 10^{-29} A^2 \Rightarrow$ which is extremely small.
Compared to the thermal noise.

Total noise

$$\sigma_{data}^2 = \sigma_1^2 + \sigma_A^2 \approx 4.9716 \times 10^{-15} + 1.11 \times 10^{-29} \\ \approx 4.9716 \times 10^{-15} A^2.$$

ASE noise is negligible because P_{opt} is too small.

Cal BER.

$$Q \approx \sqrt{SNR/2} = \sqrt{1.44} \times 10^4 / 2 \approx 120/2 \approx 60. \\ \text{Very large } Q \rightarrow BER \approx 0.$$

$$SNR_{new} = \frac{I_s^2}{\sigma_1^2}$$

$$SNR_{new} = \frac{(8.48 \times 10^{-6})^2}{(4.9716 \times 10^{-15})} \approx \frac{7.14 \times 10^{-11}}{4.9716 \times 10^{-15}}$$

$$\approx 1.436 \times 10^4$$

$$\text{linear SNR} \approx 1.44 \times 10^4 \rightarrow \text{dB: } 10 \log_{10}(1.44 \times 10^4)$$

$$\approx 41.6 \text{ dB.}$$

$$\text{BER: } Q \approx \sqrt{SNR} / 2 = \sqrt{1.44 \times 10^4} / 2 \approx 120 / 2 \approx 60$$

Very large $Q \rightarrow \text{BER} \approx 0$.

$$SNR_{new} \approx 1.44 \times 10^4 (41.6 \text{ dB}),$$

BER is 0.

4. $R = 1 \text{ A/W}$

$$\Delta f = 1 \text{ GHz}$$

$$F_n = 3$$

$$T = 300 \text{ K}$$

$$R_L = 100 \Omega$$

No pre amp $P_o = 0$

$$\sigma_s^2 = 0.$$

Compute thermal noise

$$\sigma_T^2 = F_n \frac{4k_B T \Delta f}{R_L}$$

$$4k_B T = 4 \cdot 1.381 \times 10^{-23} \times 300$$

$$\approx 1.6572 \times 10^{-20}$$

$$\Delta f \times F_n \approx 1.6572 \times 10^{-20} \times 10^9 \times 3$$

$$\approx 4.9716 \times 10^{-11}$$

$$/R_L = \frac{4.9716 \times 10^{-11}}{100}$$

$$= 4.9716 \times 10^{-13} \text{ A}^2.$$

$$\sigma_T = \sqrt{4.9716 \times 10^{-13}} \approx 7.05 \times 10^{-7} \text{ A.}$$

Minimal optical power

$$SNR = \frac{(RP_{opt})^2}{\sigma_I^2} \geq 1.$$

$$P_{opt\min} = \frac{\sigma_I}{R} \approx \frac{7.05 \times 10^{-7}}{1} \approx 7.05 \times 10^{-7} \text{ W.}$$

= 705 nW.

for $BER = 10^{-9}$

$BER \rightarrow Q$ -factor $Q \approx 6$

$$Q = \frac{RP_I}{2\sigma_I} \Rightarrow P_I = \frac{2\sigma_I Q}{R}$$

$$P_I = \frac{2 \cdot 7.05 \times 10^{-7} \cdot 6}{1} \approx 8.46 \times 10^{-6} \text{ W}$$

8.46 (μW).

The numbers are different - Why.

$SNR \geq 1$: The signal just detectable above noise.

This will still make receivers ~~errors~~ encounter errors.

$BER = 10^{-9}$: This requires much more signal so the receiver can reliably distinguish 1's & 0's minimizing errors.

5. Loss reduces the received optical power, which lowers the photocurrent while loss stays roughly the same. This would decrease the SNR & Q-factor making the '1' & '0' less distinguishable, so the BER increases.

Dispersion causes optical pulses to spread in time, overlapping adjacent bits (intersymbol interference). This would close the eye diagram and reduces the effective Q, again lowering SNR and increasing BER because the receiver cannot clearly tell a 1 from 0.